

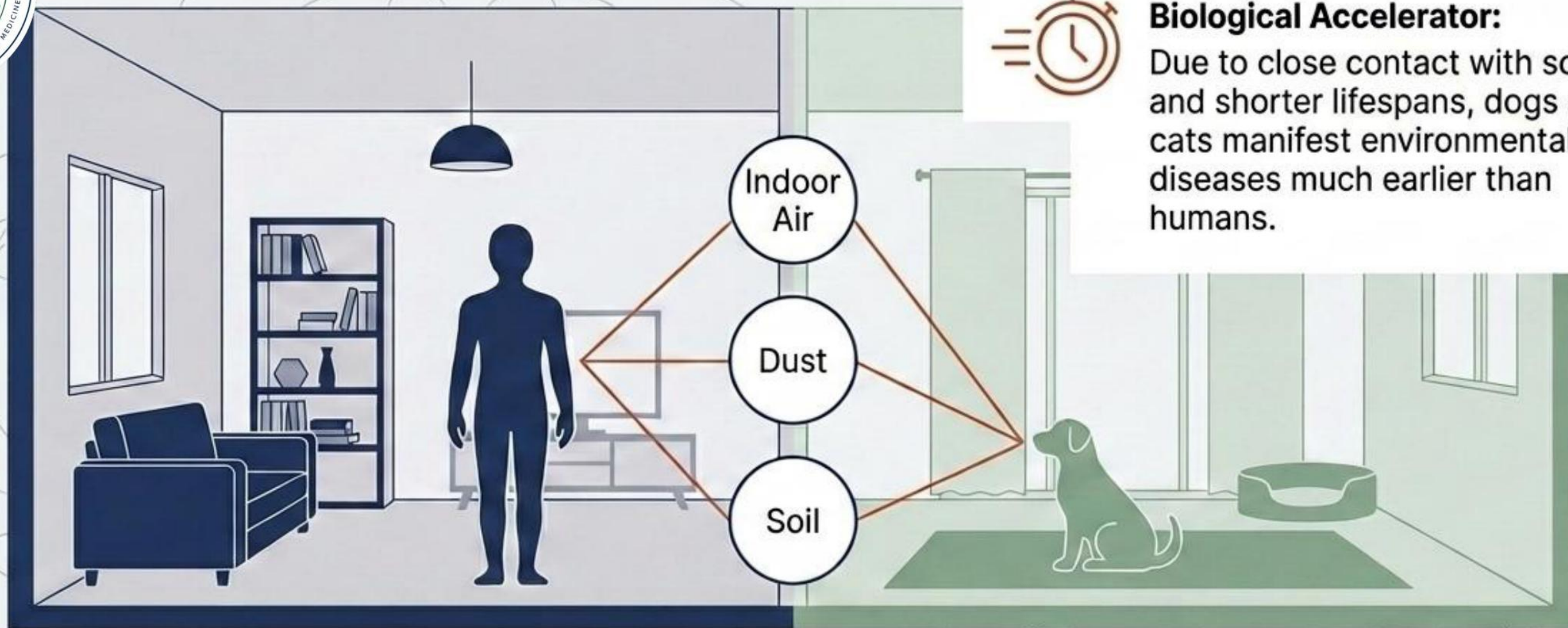


Sentinels of the Anthropocene

How animals reveal invisible threats in domestic, urban, and global environments.

The 'One Health' Approach to Modern Toxicology.





Validated Practical Example

The incidence of pleural mesothelioma in dogs is an excellent early bio-indicator of past environmental exposure to asbestos for the human population residing in the same environment.

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- [1] Glickman, L. T., Domanski, L. M., & Hayes, H. M. (1983). Canine mesothelioma: a model for human disease. *American Journal of Industrial Medicine*, 4(1), 219-227.
- [2] Backer, L. C., et al. (2001). Household and Environmental Contaminants. *Journal of Environmental Health*, 64(6),

The Historical Warning



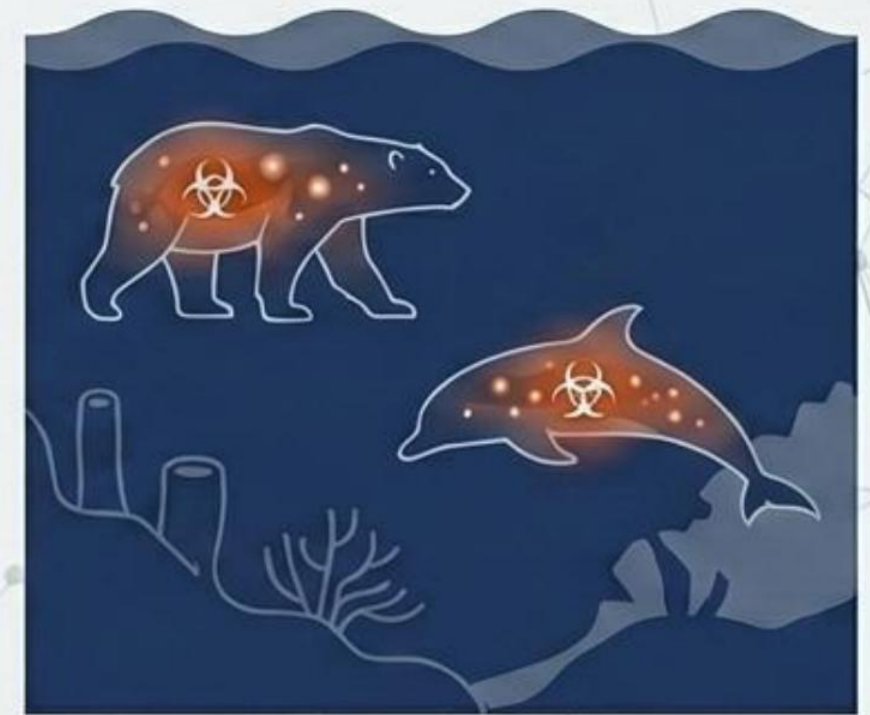
Avian eggshell thinning was the first and decisive warning signal against the overuse of the insecticide DDT.

Extreme Vulnerability



As aquatic and terrestrial species, they exhibit rapid rates of feminization and severe developmental anomalies in response to pesticide exposure.

Bioaccumulation



Enormous burdens of persistent organic pollutants (POPs) concentrate in their tissues. In mollusks, phenomena of *imposex* serve as a barometer for marine health.

3.70 MILLION

tonnes of agricultural pesticides used globally in 2022 [1].

385 MILLION

estimated cases of acute unintentional pesticide poisoning worldwide [2].

≈11,000

estimated annual deaths from acute unintentional pesticide poisoning worldwide [2].



The Ecological Cost

Intensive agriculture relies on large volumes of chemical inputs. Pollinators and other non-target organisms are among the first biological indicators of ecosystem pressure.

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[1] FAO (2023). Pesticide Use, FAOSTAT.

[2] Boedeker W. et al. (2020). The global distribution of acute unintentional pesticide poisoning: estimations based on a systematic review. BMC Public Health, 20, 1875.

The Action 40%*



Urban Pest Control:
Dispersion of tons of insecticides (e.g., Pyrethroids) in cities against flies and mosquitoes.



Collateral Damage 60%*

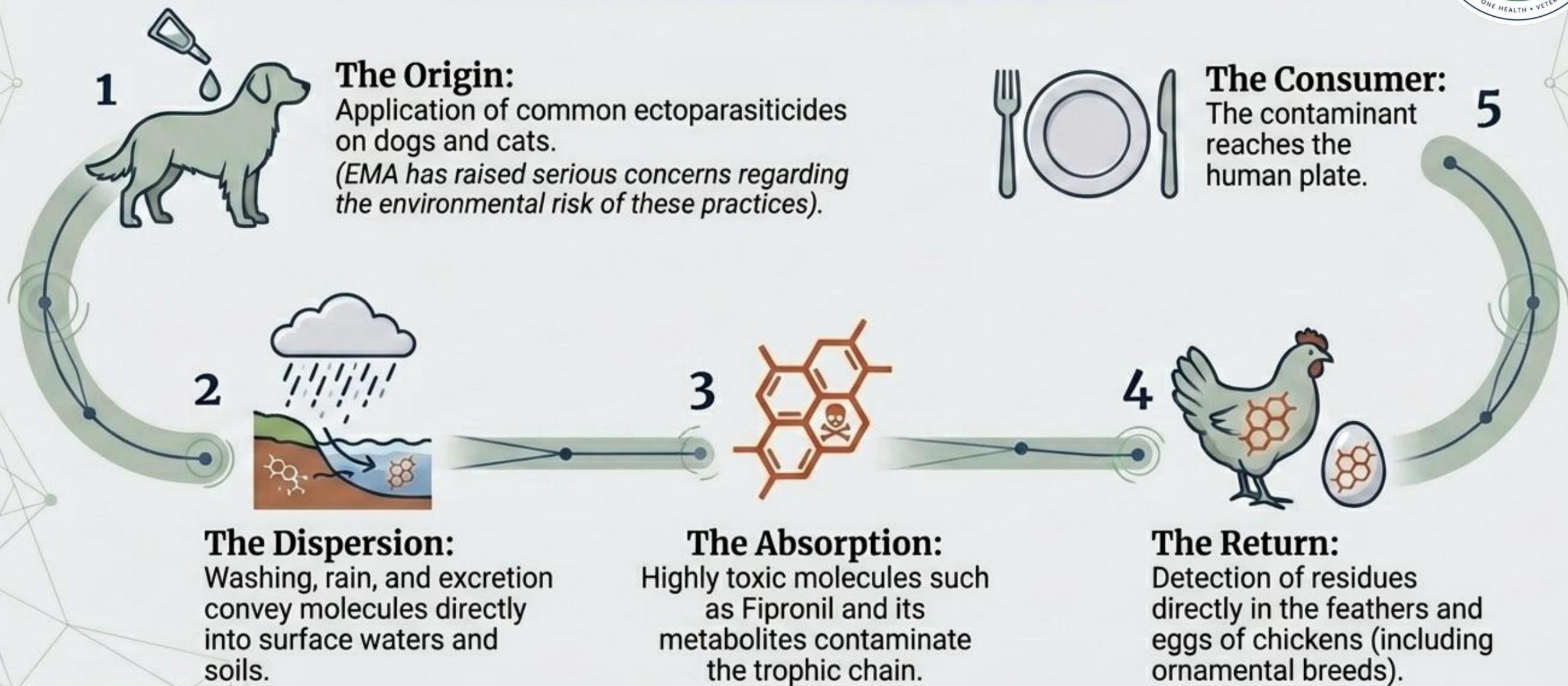


Extremely high lethal toxicity for fish and aquatic invertebrates even at minimal concentrations.



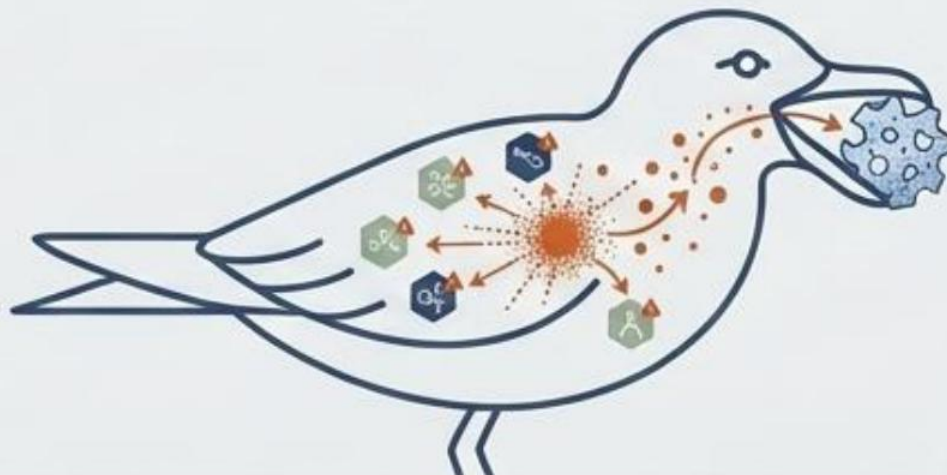
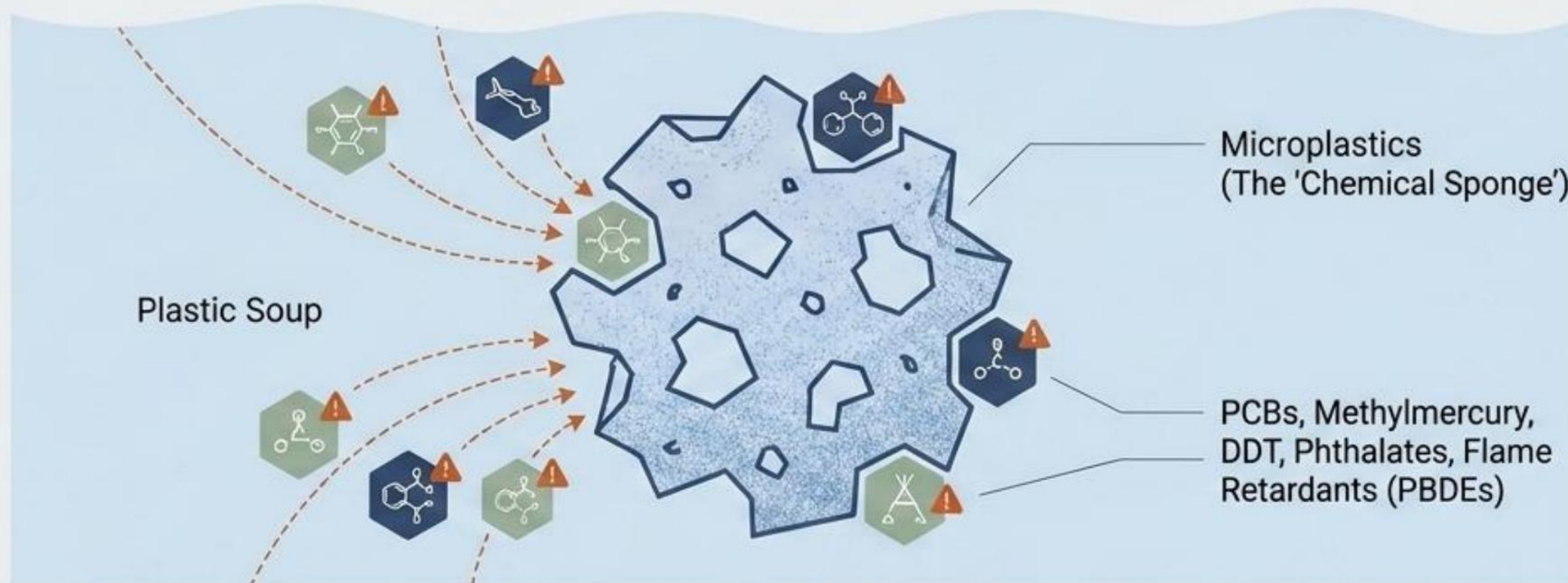
Chronic exposures to 'sub-lethal' doses of permethrin destroy olfactory memory and learning capacities of bees, compromising their survival.

Companion-animal ectoparasiticides are a relevant pathway of environmental release for active substances such as fipronil and imidacloprid.



The Cocktail Effect:

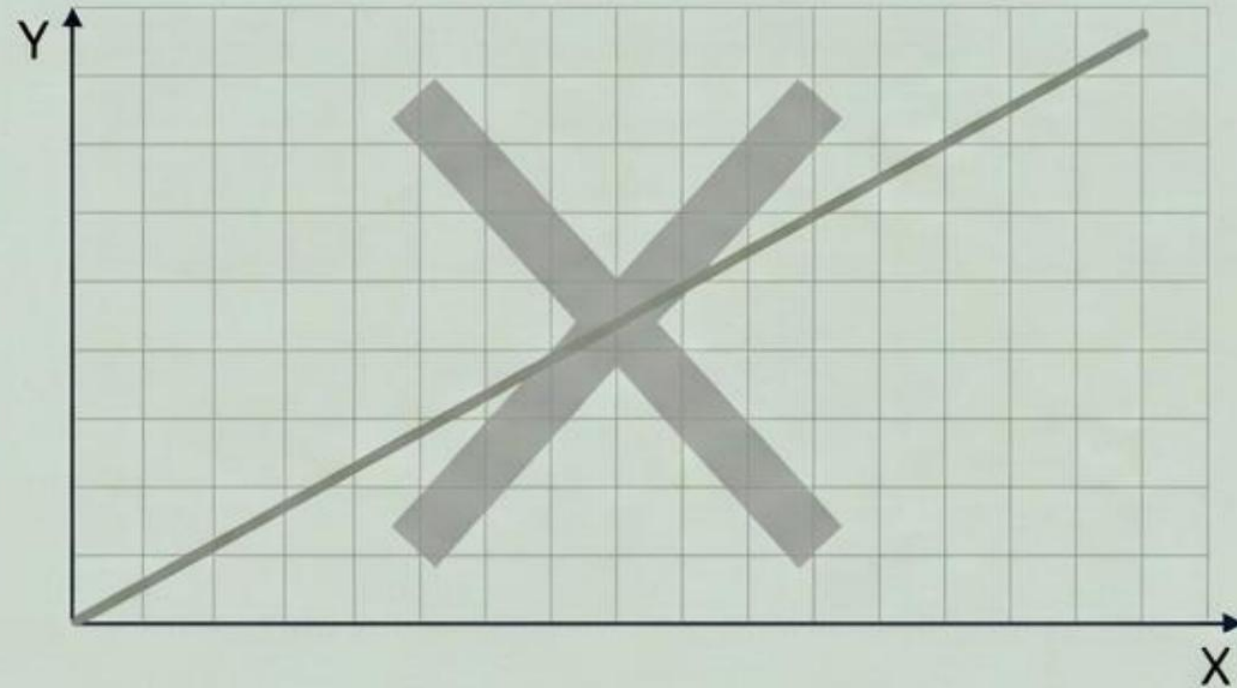
No living organism is exposed to a single isolated pollutant.
We confront countless molecules simultaneously.



Once ingested by fish or marine birds, these plastics release their toxic load directly into the animal's tissues, drastically amplifying exposure.

The Challenge of Modern Toxicology

The Old Dogma



Assessment based exclusively on lethal dosages and linear proportionality.

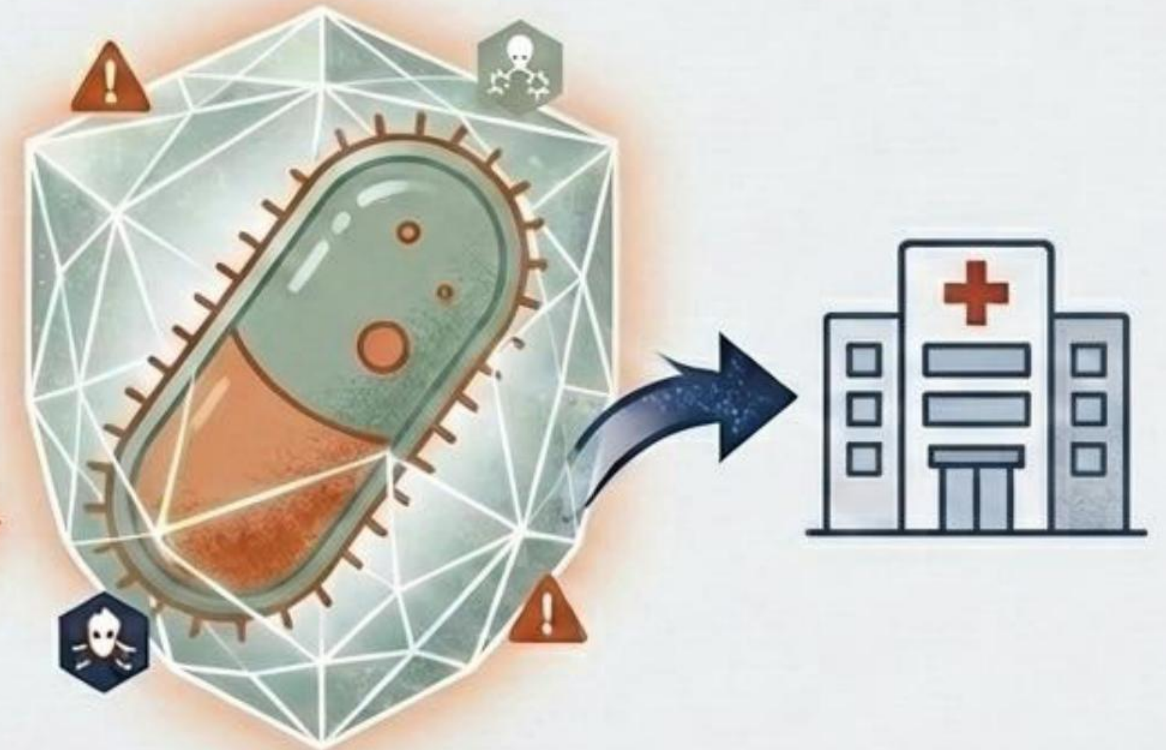
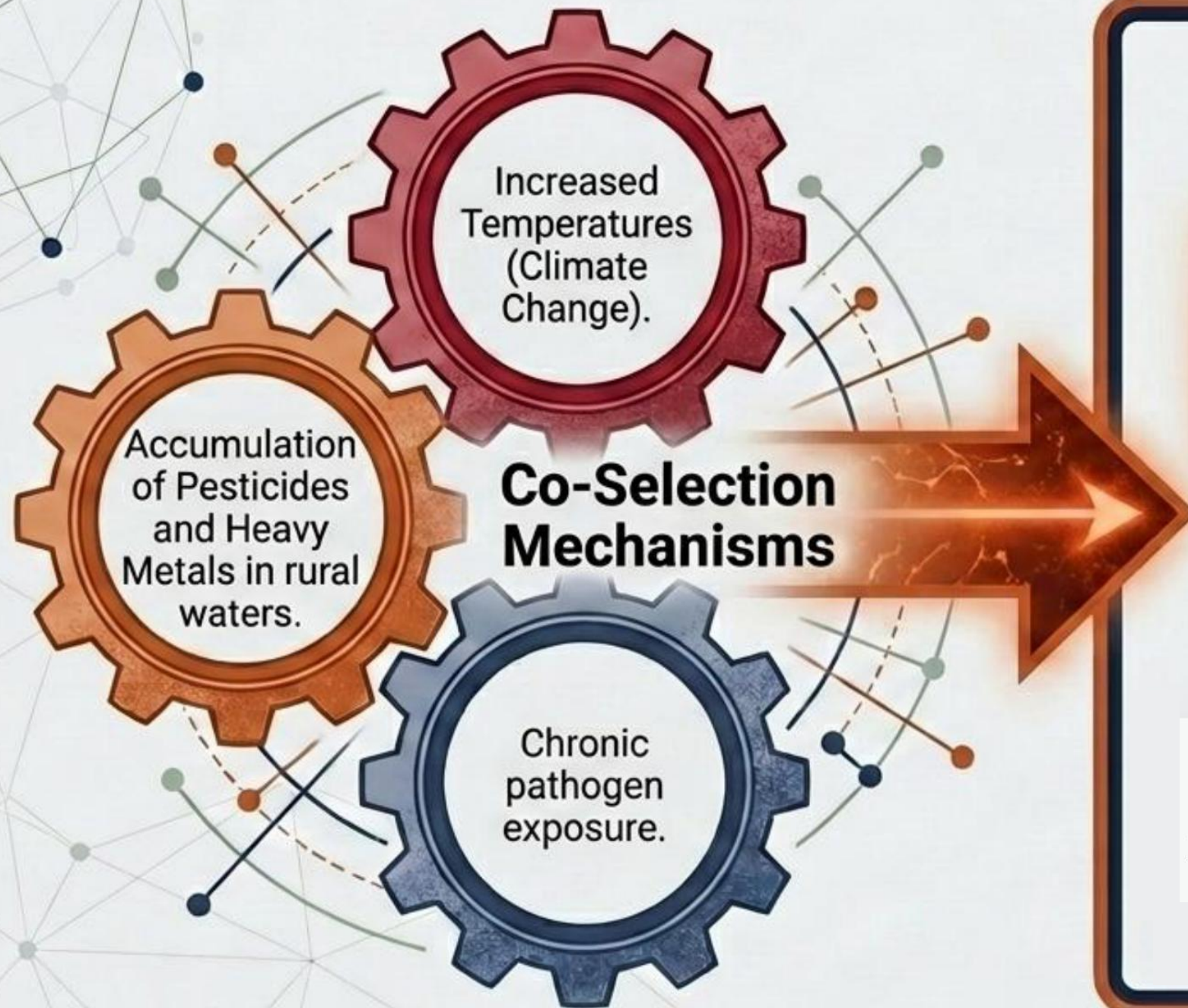
The New Reality



Non-Monotonic Curves: For endocrine disrupting chemicals (EDCs) and PFAS, effects at infinitesimal concentrations (low doses typical of chronic environmental exposure) are completely different and more subtle.

Regulatory testing should better account for low-dose, chronic, mixture-related and non-monotonic effects, particularly for EDCs and PFAS.

The Biological Equation: Co-Selection Mechanisms and AMR



Environmental co-selection

may contribute to the persistence and dissemination of antimicrobial resistance by exposing microbial communities to pesticides, heavy metals, temperature stress and chronic pathogen pressure.

The Inexorable Circularity

Pollution does not remain 'outside'. Waste, pesticides, veterinary pharmaceuticals, and PFAS released into the environment inevitably re-enter our homes through water, food, and air.



! Mixture > Single

The real risk for organisms lies in the cocktail effect. Evaluating molecules individually severely underestimates the biological hazard.

The One Health Paradigm

Pathologies in domestic animals and abnormalities in wildlife should be interpreted as potential early warning signals of shared environmental exposure and health risk.

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**Life Sciences • One Health
Veterinary Medicine**

Veterinary Toxicology
Environmental Health
Pesticides · EDCs · AMR

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